

Cross compile c-language (user space application) for raspberry pi3

Step1: Download raspberry pi toolchain from the given website

<https://github.com/raspberrypi/tools>

Apply given command in your ubuntu:

```
$ git clone https://github.com/raspberrypi/tools
```

Note: Above command will download rpi toolchain in “tools” directory

```
cd tools
```

Note: Cross compiler for cross-compiling driver as well as application is available in given folder in tools.

Step2: Find out path of cross compiler & name of cross compiler

Path of cross compiler in “tools” directory

```
/path/tools/arm-bcm2708/arm-linux-gnueabi/bin
```

Name of cross-compiler in above directory

```
arm-linux-gnueabi-gcc
```

Step3: Cross-compiling helloworld c-language program / Cross-compiling user space application

```
$ /path/tools/arm-bcm2708/arm-linux-gnueabi/bin/arm-linux-gnueabi-gcc hello.c
```

```
$ file a.out → confirm it is cross compiled
```

Step4: Connect rpi to your ubuntu host using LAN cable

Step5: Do ssh login (remote login)

```
$ ssh ip_addr_of_rpi -l pi
```

Note: Apply above command in ubuntu system .

Step6: Send cross-compiled binary (a.out) to rpi using scp command.

```
$ scp a.out pi@ip_addr:/home/pi
```

Note: Apply above command in ubuntu system.

Step7: In raspberry pi login

```
$ ./a.out
```

